



# ALIN-DANIEL RIZEA

**Associate Professor, PhD | Dean**

National University of Science and Technology POLITEHNICA Bucharest – Pitești  
University Centre, Faculty of Mechanics and Technology

## EDUCATION AND TRAINING

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**PhD in Industrial Engineering (2002)**

POLITEHNICA University of Bucharest. Thesis on machinability and electrical properties of ceramic materials.

**Specialist training in product design and additive manufacturing (2004)**

INSIC and CIRTES, Saint-Dié-des-Vosges, France.

**Research training in ceramic materials (1996–1999)**

École Centrale de Paris, SPMS Laboratory, France.

**Engineering degree in Manufacturing Technology (1987–1992)**

Polytechnic Institute of Bucharest.

**Research training (2017, 2023)**

Université de Lorraine, France.

## PROFESSIONAL EXPERIENCE

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**Dean | Faculty of Mechanics and Technology (2016–present)**

Dean at the University of Pitești through May 2023 and at POLITEHNICA Bucharest, Pitești University Centre from May 2023. Member of the university Administrative Board.

**Associate Professor | University of Pitești / POLITEHNICA Bucharest (2007–present)**

Teaches polymer and composite product manufacturing, non-metallic materials, quality management, value analysis, tolerances and dimensional control. Coordinates the Manufacturing Technology degree programme and the Quality of Products and Processes Laboratory.

**Senior Lecturer (2000–2007); Assistant Lecturer (1995–2000); Junior Lecturer (1993–1995)**

Teaching and research in manufacturing, product design, quality and non-metallic materials.

**Design Engineer | Rotan SA, Pitești (1992–1993)**

Developed engineering solutions for industrial applications.

## RESEARCH INTERESTS

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Additive manufacturing and product design; polymer and composite processing; quality engineering; ceramic materials; dimensional accuracy and artificial intelligence in manufacturing.

## SELECTED RECENT PUBLICATIONS

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**2026 Nițu, A.; Rizea, A.-D.; Iordache, D.-M.; et al.**

Liquid Metal Embrittlement Effect on the Fracture Behaviour of 316L Stainless Steel in a Liquid Lead Environment. *Metals* 16(7), 704. <https://doi.org/10.3390/met16070704>

**2025 Rizea, A.-D.; et al.**

Dimensional Accuracy Analysis of Splined Shafts and Hubs Obtained by Fused-Deposition Modeling 3D Printing Using a Genetic Algorithm and Artificial Neural Network. *Applied Sciences* 15(7), 3958. <https://doi.org/10.3390/app15073958>

**2025 Rizea, A.-D.; et al.**

The Influence of Accidental Overheating on the Microstructure and Hardness of the Inconel 718 Alloy. *Applied Sciences* 15(6), 3057. <https://doi.org/10.3390/app15063057>

## RESEARCH PROJECTS

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Partner lead in the national consortium on additive manufacturing of complex, highly stressed components (2018–2021). Led local projects on AI-assisted design of automotive components for 3D printing and selective laser melting of metal structures. Directed industry projects on SMED and TPM (2022) and injection-moulded product quality (2023).

## SCIENTIFIC OUTPUT

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The 2024 CV records 101 journal articles, 22 conference papers, 10 specialist books, eight laboratory guides, three granted patents and two patent applications.

## LANGUAGES

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Romanian: native | French: B2 | English: A1 (levels reported in the 2024 CV).

Date: September 2026